YouTube Channel Analytics

Data Analyst Report

Dataset: Youtube_Dashboard.xlsx | Period: Jan 2025 – Jul 2026 | 577 daily records

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1. Business Problem

The channel has accumulated 19 months of daily performance data covering views, watch time, subscriber growth, revenue, and audience demographics, but this data has not yet been translated into actionable insight. Content and growth decisions are currently being made without a clear read on which audiences, devices, and traffic sources actually drive performance.

This report analyzes the daily-level dataset to answer three practical questions the channel owner needs answered before planning the next content and monetization cycle:

- Which traffic sources, devices, and audience segments generate the most views, watch time, and revenue?
- Are the channel's core performance metrics — views, watch time, subscriber growth, and revenue — actually related to one another, or do they move independently?
- Is channel performance seasonal, and if so, when should content and campaigns be concentrated?

The goal is to move the channel from intuition-based decisions to a data-backed content and monetization strategy, using the exploratory analysis, correlation testing, and segment comparisons that follow.

2. Exploratory Data Analysis Insights

The dataset (‘Youtube Data’ sheet) contains 577 rows and 12 columns, spanning January 1, 2025 to July 31, 2026, with no missing values in any column. It records five numeric performance metrics — Views, Average View Duration, Subscribers Gained, Watch Time Hours, and Revenue — alongside categorical fields for Age Group, Gender, Device, Source, and Country. The sections below walk through summary statistics, a filtering example, correlation analysis, targeted research questions, and final recommendations.

a. Summary Statistics

Descriptive statistics were computed for all five numeric columns in the dataset:

Statistic	Views	Avg View Duration (min)	Subscribers Gained	Watch Time (hrs)	Revenue ($)
Count	577	577	577	577	577
Mean	2,826.13	23.90	24.93	237.11	1,238.34
Std Dev	1,234.92	11.18	16.13	122.23	682.22
Min	450.00	3.00	-5.00	20.00	7.56
25%	1,788.00	15.00	11.00	132.00	689.53
50% (Median)	2,881.00	24.00	26.00	245.00	1,232.00
75%	3,790.00	33.00	38.00	337.00	1,791.42
Max	5,249.00	45.00	76.00	449.00	2,497.00

Table 1: Summary statistics for all numeric metrics

The metrics show wide day-to-day variation — Views range from 450 to 5,249 and Revenue from about $8 to nearly $2,500 — indicating high volatility in daily performance rather than a stable baseline. Notably, Subscribers Gained has a negative minimum (-5), meaning some days saw net subscriber loss. The means and medians are close for most metrics, suggesting roughly symmetric distributions without extreme skew.

b. Data Filtering

To identify what a strong-performing day looks like, the dataset was filtered for days where Views exceeded 4,000 AND Revenue exceeded $2,000. This isolated 21 of 577 days (about 3.6% of the dataset) as the channel's top-performing days. The 10 highest-revenue days from that filtered set are shown below:

Date	Views	Revenue ($)	Device	Source	Country
2025-05-13	4,606	2,492.41	Tablet	Direct	India
2025-07-27	4,965	2,465.50	Mobile	Direct	Canada
2025-08-08	4,297	2,464.45	Mobile	Browse Features	Canada
2025-09-26	4,399	2,409.69	Mobile	YouTube Search	United Kingdom
2025-07-30	4,558	2,370.32	Desktop	Suggested Videos	Canada
2025-12-23	4,257	2,341.71	Desktop	YouTube Search	Germany
2025-06-15	4,196	2,322.53	Mobile	Browse Features	India
2025-10-30	4,578	2,309.29	Mobile	YouTube Search	Australia
2025-01-12	4,138	2,309.11	Mobile	Browse Features	United States
2025-07-14	4,942	2,298.52	Mobile	YouTube Search	India

Table 2: Top 10 days by revenue, filtered to Views > 4,000 and Revenue > $2,000

Mobile is the most common device among these top-performing days, and Direct and YouTube Search appear most frequently as traffic sources — consistent with the broader averages explored in the Research Questions section below.

c. Correlation Insights

A Pearson correlation matrix was computed across the five numeric metrics to test whether higher views translate into more watch time, subscribers, or revenue.

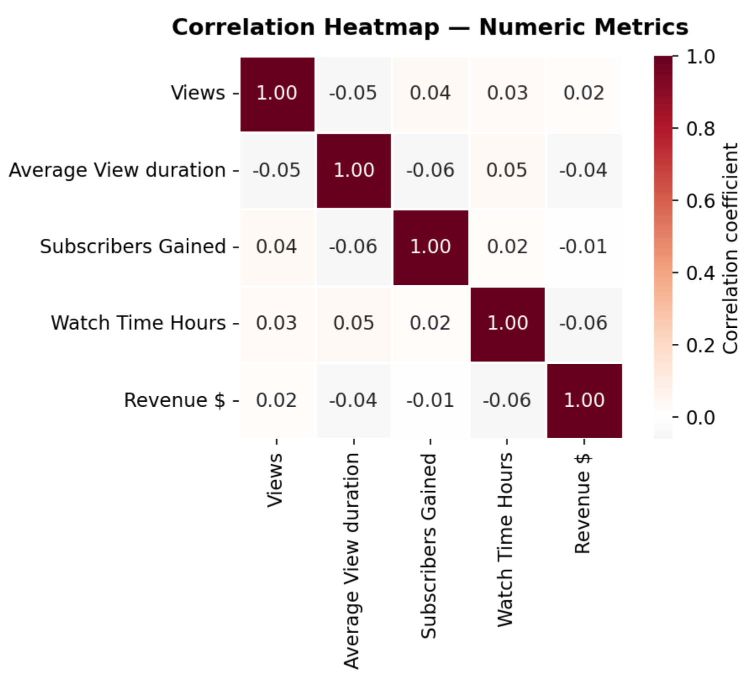

Figure 1: Correlation heatmap of Views, Average View Duration, Subscribers Gained, Watch Time Hours, and Revenue

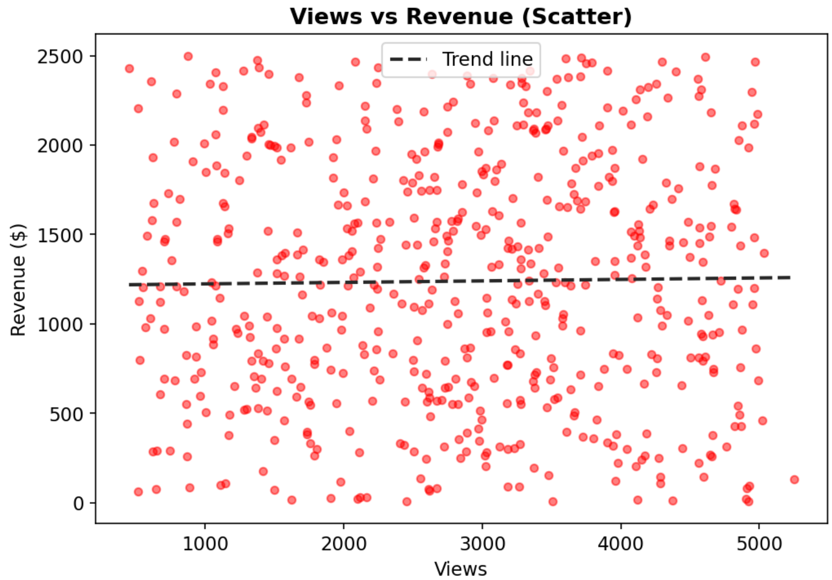

Figure 2: Views vs Revenue scatter plot with trend line

Every pairwise correlation in the matrix falls between -0.06 and 0.05 — effectively zero. Views do not meaningfully predict Watch Time, Subscribers Gained, or Revenue, and none of the other metrics correlate with each other either. The scatter plot confirms this visually: points are spread with no discernible upward or downward pattern, and the trend line is nearly flat.

This is an important finding: on this channel, a high-view day is not reliably a high-revenue or high-subscriber-growth day, and vice versa. Each metric appears to be driven by independent factors (e.g. video topic, audience segment, or ad-fill rate) rather than by view volume alone, so view count alone should not be used as a proxy for monetization or growth success.

d. Research Questions & Key Findings

Q1: Is channel performance seasonal?

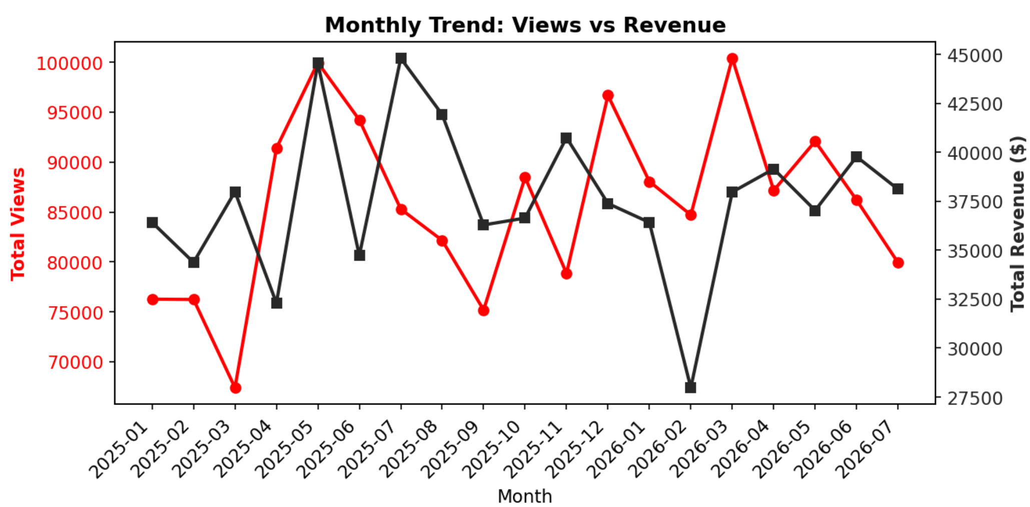

Figure 3: Monthly trend of total Views and total Revenue

Views and Revenue both show a clear seasonal dip in the second half of the year (roughly August through December) compared to the first seven months, where monthly totals are consistently higher. This points to a recurring engagement cycle rather than random noise, and suggests content or promotional pushes may be better timed for the January–July window.

Q2: Which traffic source generates the most revenue?

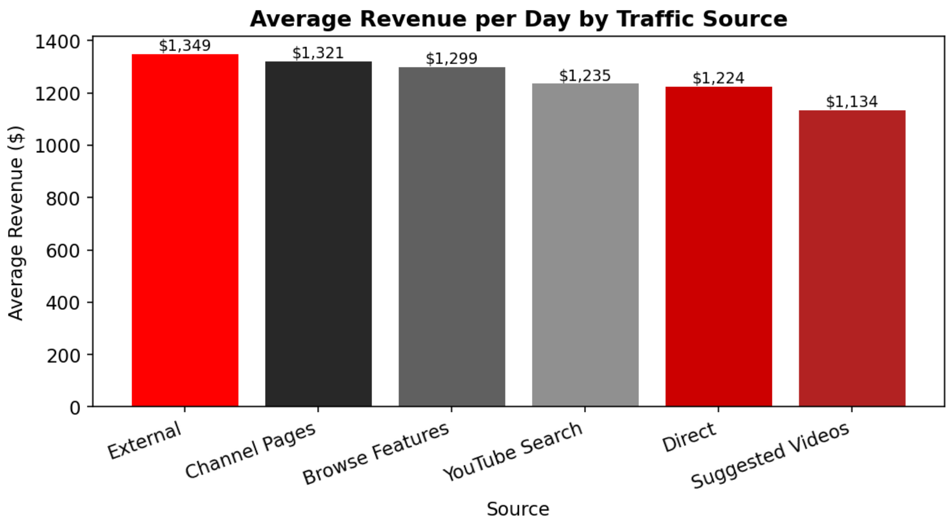

Figure 4: Average daily revenue by traffic source

External and Channel Pages traffic produce the highest average revenue per day, while Suggested Videos — despite being one of the two largest sources by volume — generates the lowest average revenue. This suggests that where a viewer comes from matters more for monetization than how many viewers a source brings.

Q3: Which age group watches the most?

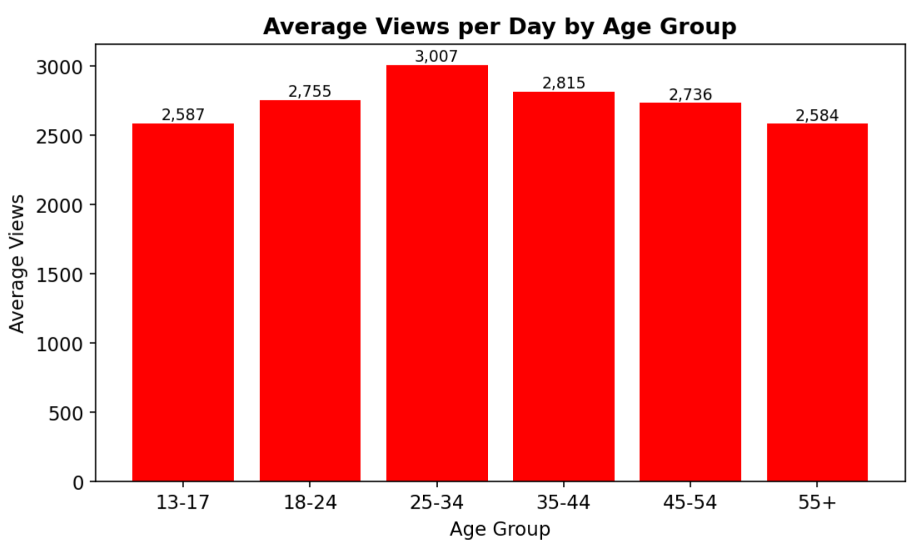

Figure 5: Average daily views by age group

The 25-34 age group generates the highest average views, followed by 35-44 and 18-24. The youngest (13-17) and oldest (55+) segments view the least on average, making 25-44 the channel's core viewing demographic.

Q4: Which device drives the most watch time?

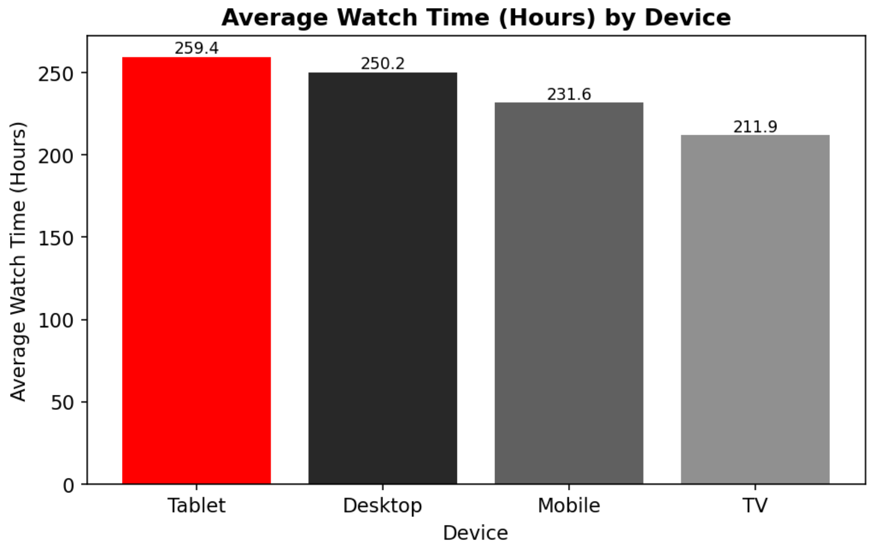

Figure 6: Average watch time (hours) by device

Tablet and Desktop sessions show the highest average watch time per day, while TV sessions — despite typically being associated with longer viewing in the industry — show the lowest average watch time in this dataset. Mobile sits in the middle on watch time but remains the most common device overall by volume (335 of 577 rows), so it still commands the largest total watch time even at a lower per-session average.

e. Final Recommendation

Based on the exploratory analysis, correlation testing, and segment-level research questions above, the following actions are recommended:

- Do not optimize purely for views. Since Views show near-zero correlation with Revenue, Watch Time, and Subscriber growth, content decisions should be evaluated against revenue and retention metrics directly, not view counts alone.
- Prioritize External traffic and Channel Pages for monetization-focused pushes, since these sources yield the highest average revenue per day, even though they aren't the largest sources by volume.
- Concentrate major content launches and campaigns in the January–July window, where both views and revenue are consistently higher, and treat August–December as a lower-intensity period.
- Continue targeting the 25-44 age group as the core audience, while testing content angles that could re-engage the under-24 and 55+ segments, which currently under-index on views.
- Invest in the mobile experience, since it is the most-used device by volume, while also optimizing for tablet and desktop viewing, which post the highest average watch time per session; TV underperforms on both counts and may need a distinct content or promotion strategy if it is to be grown.
- Investigate the days with negative Subscribers Gained (34 of 577 days) to understand whether specific content, sources, or external events are driving unsubscribes, since this pattern is currently unexplained by any other metric in the dataset.

Together, these steps shift the channel's strategy from a views-first mindset to a revenue- and retention-aware approach, grounded in the segments and time periods that the data shows actually perform.